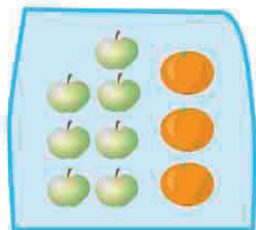


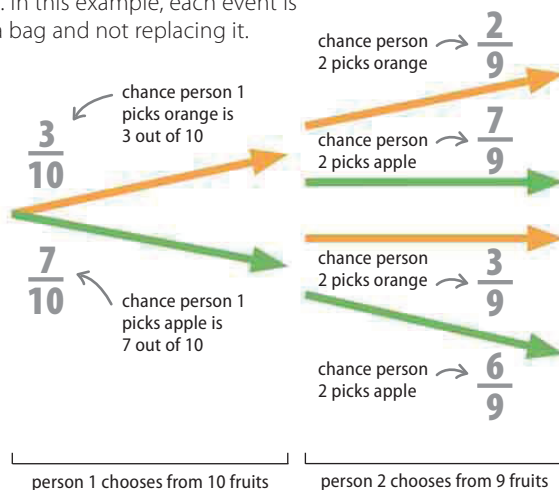
When multiple events are dependent

Tree diagrams show how the chances of one event can depend on the previous event. In this example, each event is someone picking a fruit from a bag and not replacing it.



△ Dependent events

The first person picks from a bag of 10 fruits (3 oranges, 7 apples). The next picks from 9 fruits, when the chances of what is picked are out of 9.



Find the probability

What are the chances that the first and second person will each choose an orange? Multiply the chances of both events together.

$$\frac{3}{10} \times \frac{2}{9} = \frac{6}{90}$$

chance person 1 picks orange

chance person 2 picks orange

or

chance both pick orange

fraction $\frac{6}{90}$ reduced down to $\frac{1}{15}$ by dividing 6 and 90 by 6

